

Binomial Distributions Worksheet

1. Identify which of the following experiments may be modelled using a binomial random variable. If so, identify the variable.
 - a. Jacob is recording the number of white cars that drive past him every minute.
 - b. A die is thrown until a 6 is obtained. The number of throws is recorded. This experiment is repeated 20 times.
 - c. A bag contains 3 red marbles and 2 black marbles. Daniel picks a marble at random, records its colour and returns it to the bag. This experiment is repeated 10 times and the total number of red marbles is recorded.
 - d. John runs 3 km every day for 30 days in a row. He records the total number of days he runs 3 km in under 10 minutes.
2. Find the mean and standard deviation for each binomial random variable below (2 dp)
 - a. $\text{Bin}(50, 0.1)$
 - b. $\text{Bin}(30, 0.15)$
 - c. $\text{Bin}(400, 0.7)$
3. For each of the following situation, calculate the mean, variance and standard deviation
 - a. A coin is tossed 10 times and X is the number of tails that are face up
 - b. A die is thrown 5 times and X is the number of 1 or 3 that occurs.
 - c. Two coins are tossed 10 times and X is the number of times two tails are facing up.
4. A newly opened local business is having a promotion. Every day, the first 10 customers to purchase something can enter the prize draw to decide 1 winner. Becky entered the prize draw 9 days in a row but only won once. Find the mean and standard deviation for the number of wins. Is Becky luckier or unluckier than expected?
5. A player throws 12 dice, let X be the number of 1s. Find the expected value and standard deviation of X . What is the probability of getting exactly the expected number of 1s? (2 dp)
6. At a local TV game show, the contestant needs to answer 48 questions in succession and receives prize money based on the number of questions they answer correctly. Each question has 4 options. Bobby decided to randomly guess each question for comedic effect. What is the mean and standard deviation of the random variable of the number of question Bobby answers correctly?

In fact, Bobby got 18 questions correct and left with \$180. How many standard deviations above or below the mean is his performance?
7. Bobby is qualified for the national TV game show, where he now needs to answer 100 questions in succession. Each question has 5 options to discourage guessing. Undeterred by this, Bobby decided to randomly guess again. This time, he got 32 questions correct and left with \$3200. Is his performance more impressive compared to the local competition? Use mean and standard deviations to explain.

8. It is known that only 20% of people with a driver's licence actually understands road rules. Two teams are attempting to improve this percentage by implementing educational programs. Both conduct random tests at the end of the program. Team A runs a trial with 40 drivers, of whom 10 passed the test. Team B runs a trial with 60 drivers, of whom 18 passed the test. Using the mean and standard deviation of each trial to decide which team is more effective at educating drivers.
9. It is known that 10% of people are lactose intolerant. Robert and Betty are hosting their wedding where a gigantic cheese fountain will be featured. 80 guests are invited to their wedding. (2 dp)
- What is the expected value and standard deviation for the number of lactose intolerant guests?
 - What is the probability that among the 80 guests, the number of lactose intolerant guests lies within $\frac{1}{3}$ standard deviation from the mean?
10. Let $X \sim \text{Bin}(n, p)$ where n is a fixed constant. Find σ^2 in terms of p and n . Hence, find the value of p and σ^2 when σ^2 is maximised.
11. If a Bernoulli trial is repeated until the first success is obtained, the total number of trials taken then follows what is called a Geometric distribution. For each Bernoulli trial, the probability of success is p and $q = 1 - p$ for failure. X is the number of trials until the first success, i.e. X is a positive integer.
- Show that $P(X = x) = pq^{x-1}$
 - Show that $\mu = p + 2pq + 3pq^2 + \dots$
 - Find an expression for $\mu - q\mu$
 - Hence find an expression for μ

Solutions:

Q1 a: no, more than two outcomes, b: no, more than two outcomes, c: yes, number of red marbles, d: no, not independent as John is likely to improve his time with practice.

Q2 a: $\mu = 5, \sigma = 2.12$, b: $\mu = 4.5, \sigma = 1.96$, c: $\mu = 280, \sigma = 9.17$

Q3 a: $\mu = 5, \sigma^2 = 2.5, \sigma = \sqrt{\frac{5}{2}}$, b: $\mu = \frac{5}{3}, \sigma^2 = \frac{10}{9}, \sigma = \frac{\sqrt{10}}{3}$, c: $\mu = 2.5, \sigma^2 = \frac{15}{8}, \sigma = \sqrt{\frac{15}{8}}$.

Q4: $\mu = \frac{9}{10}, \sigma = \frac{9}{10}$, she is luckier than expected. Q5: $\mu = 2, \sigma = 1.29, P = 0.30$. Q6: $\mu = 12, \sigma = 3$, 2 standard deviations over. Q7: $\mu = 20, \sigma = 4$, more impressive this time as it's 3 standard deviations over compared to 2. Q8: $\mu_A = 8, \sigma_A = 2.53, \mu_B = 12, \sigma_B = 3.10$, team A is 0.79σ over while team B is 1.94σ over. Therefore, team B is more effective. Q9: $\mu = 8, \sigma = 2.68, P =$

0.147. Q10: $\sigma^2 = np - np^2, p = \frac{1}{2}, \sigma = \frac{n}{4}$.

Q11 c: $\mu - q\mu = p + pq + pq^2 + \dots$, d: $\mu = \frac{1}{p}$